

# Robust Bayesian Analysis with information geometry and Perturbed-Law based sensitivity Indices

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# Introduction

- ▶ **Bayesian inference** provides a comprehensive probabilistic framework for quantifying epistemic uncertainties and updating them from new information, typically measured data  
(Gelman et al., 2013; Kaipio and Somersalo, 2005)
- ▶ It is based on the definition of a **prior distribution**, which summarizes a state of knowledge about some input parameters before making any observation
- ▶ By conditioning on observed data, Bayesian inference enables to derive a **posterior distribution**, reflecting an updated state of knowledge about parameters
- ▶ In this framework, the selection of the prior distribution is of utmost importance !  
⇒ Necessity to analyze the impact of the choice of the prior on Bayesian inference results

# Introduction

- ▶ **Robust Bayesian analysis** aims at studying the sensitivity of Bayesian inference outcomes with respect to the inputs for that inference (including priors) (Berger, 1990; Ruggeri et al., 2005)
  - ▶ More recently, the emerging field of **robustness analysis** enables the development of methods measuring the impact of the choice of an input distribution, by studying variations of a quantity of interest (QoI) with respect to this choice (Gauchy et al., 2022; Iooss et al., 2020)
  - ▶ In particular, Perturbed-Law based sensitivity Indices (PLI) are interesting tools to study variations of a given QoI, for a given perturbation of the input distribution (Lemaître et al., 2015; Sueur et al., 2016)
- ⇒ **Main goal: definition of PLI dedicated to Bayesian inference**

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# General framework of Bayesian inference

- ▶ Let  $\mathbf{X}, \mathbf{Y}$  be random variables respectively taking values in  $\mathcal{X} \subset \mathbb{R}^d$  and  $\mathcal{Y} \subset \mathbb{R}^n$ , where:
  - $\mathbf{X}$  represents uncertain input **parameters** of a given model,
  - $\mathbf{Y}$  represents **observables**.
- ▶ Input parameters  $\mathbf{X}$  are assumed to admit a density  $\pi_{\mathbf{X}}$ , known as the **prior** density
- ▶ Given observed data  $\mathbf{y} \in \mathcal{Y}$ , **Bayes' theorem** enables to derive the following expression for the **posterior** density (i.e., the conditional density of  $\mathbf{X}$  knowing  $\mathbf{Y} = \mathbf{y}$ ):

$$\pi_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) = \frac{\mathcal{L}(\mathbf{x})\pi_{\mathbf{X}}(\mathbf{x})}{C}$$

where:

- $\mathcal{L}(\mathbf{x}) = \pi_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x})$  is the **likelihood** (i.e., the conditional density of  $\mathbf{Y}$  knowing  $\mathbf{X} = \mathbf{x}$ , for  $\mathbf{x} \in \mathcal{X}$ ),
- $C > 0$  is the **model evidence**:

$$C = \int_{\mathcal{X}} \mathcal{L}(\mathbf{x})\pi_{\mathbf{X}}(\mathbf{x})d\mathbf{x}.$$

# Bayesian inference seen as a reliability problem: the BUS framework

- ▶ Bayesian inference can be reformulated into a *structural reliability* problem, via the **BUS framework** (*Bayesian Updating with Structural reliability methods*) (Straub and Papaioannou, 2015)

- ▶ Let  $\mathcal{D}_f \subset \mathcal{X} \times [0, 1]$  be the BUS *failure domain* defined by:

$$\mathcal{D}_f = \{(\mathbf{x}, v) \in \mathcal{X} \times [0, 1] \mid v \leq \beta \mathcal{L}(\mathbf{x})\}$$

where  $\beta > 0$  is a constant such that  $\beta \mathcal{L}(\mathbf{x}) \leq 1$ , for all  $\mathbf{x} \in \mathcal{X}$ .

- ▶ Given  $\Upsilon \sim \mathcal{U}([0, 1])$  independent from  $\mathbf{X}$ , a realization  $(\mathbf{x}, v) \in \mathcal{X} \times [0, 1]$  of  $(\mathbf{X}, \Upsilon)$  lies in  $\mathcal{D}_f$  if and only if  $\mathbf{x}$  follows the posterior distribution
- ▶ The BUS **failure probability** is linked to model evidence  $C$  :

$$P_f = \mathbb{P}((\mathbf{X}, \Upsilon) \in \mathcal{D}_f) = \int_0^1 \int_{\mathcal{X}} \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}, v) \pi_{\mathbf{X}}(\mathbf{x}) d\mathbf{x} dv = \beta C$$

- ▶ Recent development of structural reliability methods tailored to BUS framework, in order to sample from the posterior distribution (e.g., subset simulation) (Betz et al., 2018; DiazDelaO et al., 2017)



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# Fisher distance between densities

- ▶ The prior distribution is assumed to admit independent marginals, and the associated densities  $\pi_{X_1}, \dots, \pi_{X_d}$  are assumed to belong to the family of **parametric densities**  $\mathcal{S} = \bigcup_{\Theta \subset \mathbb{R}^r, r \geq 1} \{\pi_{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \in \Theta\}$
- ▶ Given a **parametric space**  $\Theta \subset \mathbb{R}^r$ , with  $r \geq 1$ , a set of parameters  $\boldsymbol{\theta} \in \Theta$  determines a unique density  $\pi_{\boldsymbol{\theta}}$  in the manifold  $\mathcal{S}_{\Theta} = \{\pi_{\tilde{\boldsymbol{\theta}}} \mid \tilde{\boldsymbol{\theta}} \in \Theta\}$
- ▶  $\mathcal{S}_{\Theta}$  defines a Riemannian manifold when equipped with the **Fisher metric** defined by: (Rao, 1945)

$$\mathcal{I}(\boldsymbol{\theta}) = \mathbb{E} [(\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\mathbf{X})) (\nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(\mathbf{X}))^{\top}]$$

- ▶ The **Fisher distance** between two densities  $\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2} \in \mathcal{S}_{\Theta}$  is then defined by:

$$d_F(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2}) = \min_{\gamma \in \mathcal{P}(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2})} \ell(\gamma)$$

where  $\mathcal{P}(\pi_{\boldsymbol{\theta}_1}, \pi_{\boldsymbol{\theta}_2})$  is the set of paths joining  $\pi_{\boldsymbol{\theta}_1}$  to  $\pi_{\boldsymbol{\theta}_2}$ , and  $\ell(\gamma)$  the **length** (in Fisher metric sense) of  $\gamma$

# Perturbation of prior densities

- ▶ A **geodesic** is a path which minimizes its length (in the Fisher metric sense) (Costa et al., 2015)

- ▶ Given  $\pi_{\theta_0} \in \mathcal{S}_\Theta$  and  $\theta_0 \in \Theta$ , the **Fisher sphere** of center  $\pi_{\theta_0}$  and radius  $\delta > 0$  is defined by:

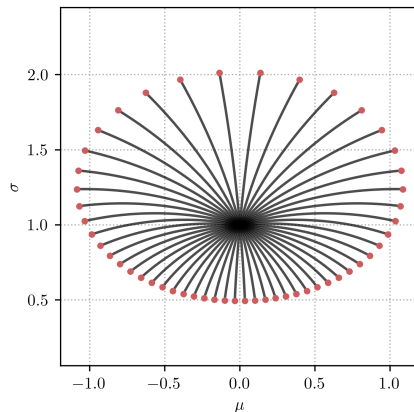
$$\partial\mathcal{B}_F(\pi_{\theta_0}, \delta) = \{\pi_\theta \in \mathcal{S}_\Theta \mid d_F(\pi_{\theta_0}, \pi_\theta) = \delta\}$$

*i.e.*, the collection of the endpoints of all geodesics starting from  $\pi_{\theta_0}$  with length  $\delta$

- ▶ Definition of a perturbed density: given a marginal  $\pi_{X_i}$  and a perturbation level  $\delta > 0$ , a perturbed density  $\pi_{X_i, \delta}$  is defined by a point lying on the Fisher sphere  $\partial\mathcal{B}_F(\pi_{X_i}, \delta)$  (Gauchy et al., 2022)
- ▶ Perturbed densities can be determined by computing geodesics, through solving a system of ordinary differential equations (Hamilton's equations) (Deriglazov, 2010)

# Perturbation of prior densities - Example of Gaussian densities

- Example of the manifold of univariate Gaussian densities  $\mathcal{S}_\Theta = \{\mathcal{N}(\cdot; \mu, \sigma^2) \mid (\mu, \sigma) \in \Theta\}$ , with  $\Theta = \mathbb{R} \times \mathbb{R}_{>0}$ :



*Samples on the Fisher sphere with center  $(0, 1)$  and radius  $\delta = 1$  (red dots), and associated geodesics (black lines)*

# Perturbed prior and posterior densities

- ▶ Definition of a perturbed prior joint density, given  $i \in \{1, \dots, d\}$  and a perturbation level  $\delta > 0$ :

$$\pi_{i\delta}(\mathbf{x}) = \left( \prod_{1 \leq j \leq d, j \neq i} \pi_{X_j}(x_j) \right) \pi_{X_{i,\delta}}(x_i),$$

where  $\pi_{X_{i,\delta}} \in \partial\mathcal{B}_F(\pi_{X_i}, \delta)$  is a perturbation of the marginal density  $\pi_{X_i}$

- ▶ The **perturbed posterior** density can then be written as follows:

$$\pi_{i\delta}(\mathbf{x}|\mathbf{y}) = \frac{\mathcal{L}(\mathbf{x})\pi_{i\delta}(\mathbf{x})}{C_{i\delta}}$$

where  $C_{i\delta} > 0$  denotes the **perturbed model evidence**:

$$C_{i\delta} = \int_{\mathcal{X}} \mathcal{L}(\mathbf{x})\pi_{i\delta}(\mathbf{x})d\mathbf{x}$$

# Perturbed Law-based Indices for Bayesian inference

- Definition of **Bayesian Perturbed Law-based Indices** (BPLI), given a perturbation  $\pi_{X_{i,\delta}} \in \partial\mathcal{B}_F(\pi_{X_i}, \delta)$ :

$$S(\pi_{X_{i,\delta}}) = \frac{C_{i\delta} - C}{C}$$

- Definition of minimal and maximal BPLI over the Fisher sphere:

$$S_{i\delta}^- = \min_{\pi \in \partial\mathcal{B}_F(\pi_{X_i}, \delta)} S(\pi) \qquad S_{i\delta}^+ = \max_{\pi \in \partial\mathcal{B}_F(\pi_{X_i}, \delta)} S(\pi)$$

- By using the BUS framework, BPLI can be reformulated into PLI for failure probabilities:

$$S(\pi_{X_{i,\delta}}) = \frac{P_{i\delta} - P_f}{P_f}$$

where  $P_{i\delta}$  denotes the following **perturbed BUS failure probability**:

$$P_{i\delta} = \int_0^1 \int_{\mathcal{X}} \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}, v) \pi_{i\delta}(\mathbf{x}) d\mathbf{x} dv = \beta C_{i\delta}$$

⇒ **Allows the use of estimation techniques dedicated to PLI for failure probabilities**

(Lemaître et al., 2015; Sueur et al., 2016)

# Practical computation of BPLI

## BPLI estimation based on BUS reformulation and reverse importance sampling

**Input:** prior density  $\pi_{\mathbf{X}}$ , marginal index  $i \in \{1, \dots, d\}$ , perturbation level  $\delta$ , Fisher sphere sample size  $K \geq 1$ , prior samples  $\{\mathbf{x}^{(k)}\}_{1 \leq k \leq N} \sim \pi_{\mathbf{X}}$  and  $\{v^{(k)}\}_{1 \leq k \leq N} \sim \mathcal{U}([0, 1])$ , failure probability estimate  $\hat{P}_{f,N}$

1. Draw  $K$  samples  $\{\pi_{X_{i,\delta}}^{(k)}\}_{1 \leq k \leq K}$  on the Fisher sphere  $\partial \mathcal{B}_F(\pi_{X_i}, \delta)$

2. **For**  $k \in \{1, \dots, K\}$ :

2.1 Estimate the perturbed BUS failure probability  $\hat{P}_{i\delta,N}^{(k)}$  associated to  $\pi_{X_{i,\delta}}^{(k)}$  :

$$\hat{P}_{i\delta,N}^{(k)} = \frac{1}{N} \sum_{l=1}^N \mathbf{1}_{\mathcal{D}_f}(\mathbf{x}^{(l)}, v^{(l)}) \frac{\pi_{X_{i,\delta}}^{(k)}(x_i^{(l)})}{\pi_{X_i}(x_i^{(l)})}$$

2.2 Estimate the PLI  $\hat{S}_N(\pi_{X_{i,\delta}}^{(k)}) = (\hat{P}_{i\delta,N}^{(k)} - \hat{P}_{f,N}) / \hat{P}_{f,N}$

3. Set  $\hat{S}_{i\delta,N}^- = \min_{1 \leq k \leq K} \hat{S}_N(\pi_{X_{i,\delta}}^{(k)})$  and  $\hat{S}_{i\delta,N}^+ = \max_{1 \leq k \leq K} \hat{S}_N(\pi_{X_{i,\delta}}^{(k)})$

**Output:** BPLI estimates  $\hat{S}_{i\delta,N}^-, \hat{S}_{i\delta,N}^+$

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# Case of a linear Gaussian model

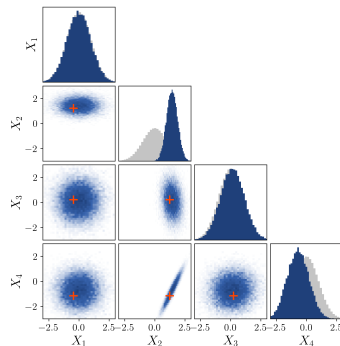
- ▶ Case of a Bayesian inverse problem with input dimension  $d = 4$ , a standard normal prior  $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$  and a linear forward model  $\mathcal{M}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ , with  $\mathbf{a} = (10^{-6}, 1, 0.1, -0.5)^\top$

- ▶ Likelihood based on an additive Gaussian model with noise  $\sigma_0 = 0.1$ :

$$Y|\mathbf{X} = \mathbf{x} \sim \mathcal{N}(\mathcal{M}(\mathbf{x}), \sigma_0^2)$$

- ▶ Given data  $y \in \mathbb{R}$ , the posterior distribution is  $\mathcal{N}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y)$ , where:

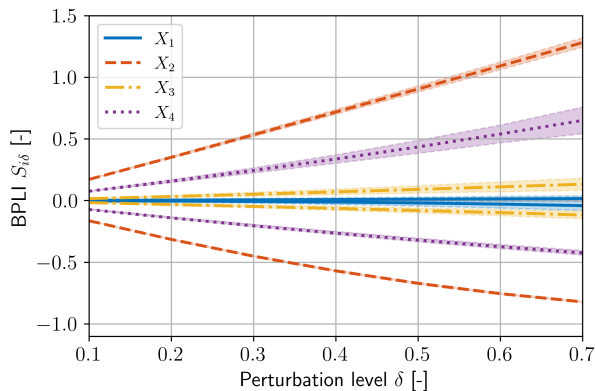
- $\boldsymbol{\mu}_y = (\boldsymbol{\Sigma}_y \mathbf{a} \sigma_0^{-2}) y$  denotes the posterior mean
- $\boldsymbol{\Sigma}_y = (\mathbf{I}_d + \sigma_0^{-2} \mathbf{a} \mathbf{a}^\top)^{-1}$  the posterior covariance matrix



Posterior distribution (in blue) obtained for the reference prior (in grey) - Samples drawn via Subset Simulation algorithm of (Betz et al., 2018)

# Case of a linear Gaussian model - BPLI results

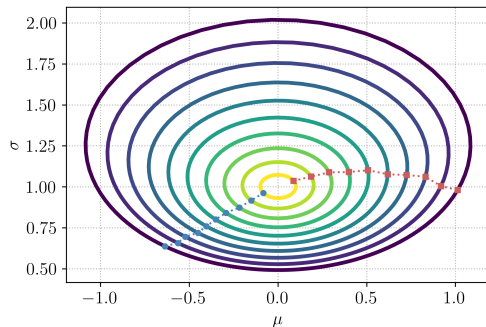
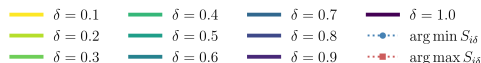
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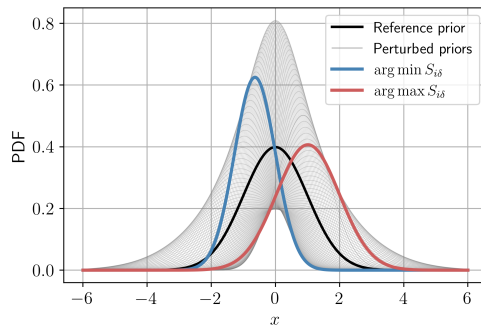
*Minimum and maximum of BPLI over the Fisher sphere of radius  $\delta$*

# Case of a linear Gaussian model - BPLI results

- Case of a Bayesian inverse problem with input dimension  $d = 4$ , a standard normal prior  $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$  and a linear forward model  $\mathcal{M}(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$ , with  $\mathbf{a} = (10^{-6}, 1, 0.1, -0.5)^\top$



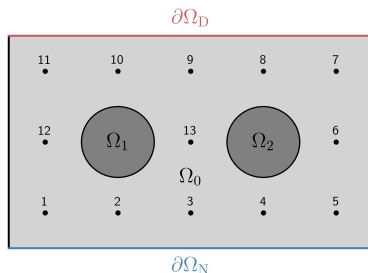
*Fisher spheres and BPLI extrema arguments*



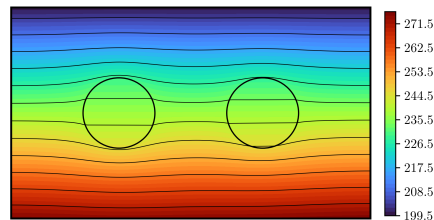
*Perturbed prior marginals densities of  $X_2$*

# Inverse heat conduction problem

- ▶ Case of an **inverse heat conduction problem**, involving the inference of thermal conductivities  $\kappa_1, \kappa_2$  of two circular inclusions in a rectangular domain, based on temperature measurements (Nagel and Sudret, 2016)
- ▶ Forward model based on 2D heat equation solved with the finite element method
- ▶ Two cases considered for observed data: (a) sensors 1, 2, 10, 11 and 12; (b) all available sensors



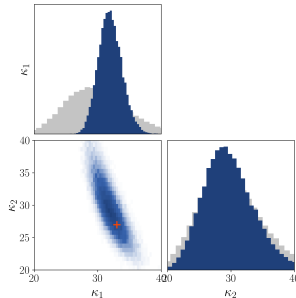
*IHCP domain and sensors location*



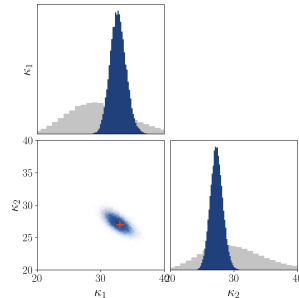
*Example of a temperature field*

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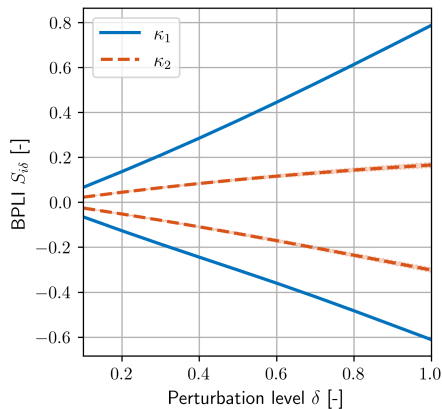
Reference posterior distribution - Case (a)



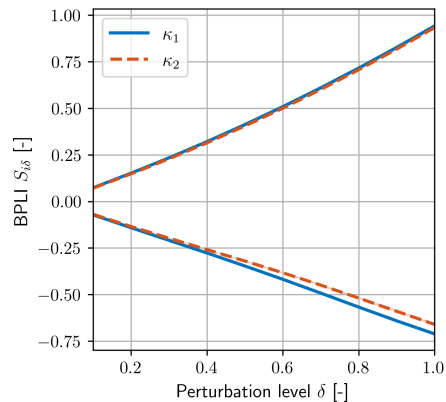
Reference posterior distribution - Case (b)

# Inverse heat conduction problem - BPLI results

- Computation of minimum and maximum of BPLI over the Fisher sphere, for each case:



Case (a) (sensors around inclusion 1 only)



Case (b) (all sensors)

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# Conclusion

- ▶ Proposition of **Bayesian Perturbed Law-based Indices (BPLI)**, aiming at quantifying the impact of the choice of the prior distribution on results of Bayesian inference
- ▶ The proposed BPLI can be reformulated into PLI for failure probabilities, for which estimation techniques have been proposed in the recent literature (Lemaître et al., 2015; Sueur et al., 2016)
- ▶ BPLI can help to reduce the dimensionality of Bayesian inverse problems, by focusing on important input parameter
- ▶ Their computation remains tractable to problems involving nonlinear forward models with possibly high-dimensional inputs, while allowing an arbitrary perturbation level for the prior  
⇒ contrasts with a significant part of existing Robust Bayesian Analysis approaches (Betrò and Guglielmi, 2000)



# Perspectives

- ▶ The likelihood ratio of reverse importance sampling can locally take high values, which can impede the quality of estimated BPLI and their confidence intervals (Gauchy et al., 2022)
- ▶ Similar problems in the case of small BUS failure probabilities (*i.e.* strongly informative data and/or high dimensional input distributions)  
⇒ need for more efficient sampling techniques for estimating perturbed BUS failure probabilities (Au and Beck, 2001; Betz et al., 2018)
- ▶ BPLI estimation cost can be prohibitive in the case of costly forward models ⇒ use of **surrogate modeling** techniques (either *prior-based* or *adaptive*) to ensure a feasible computational cost (D. R. et al., 2022; Marzouk and Najm, 2009; Wagner et al., 2021)

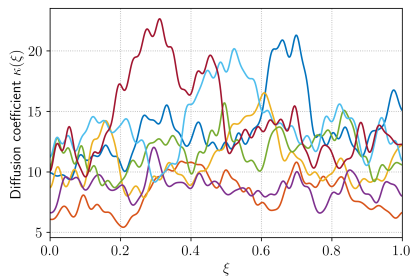
Thank you for your attention !

# Appendices

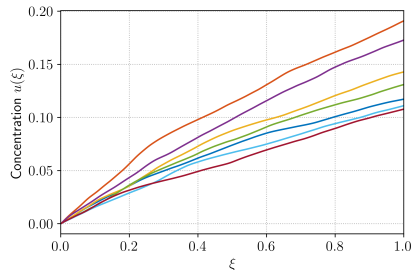
High-dimensional diffusion inverse problem

# Case of a high-dimensional diffusion inverse problem

- ▶ Case of a **high-dimensional diffusion inverse problem**, involving the identification of a diffusion coefficient random field, given concentration measurements (Fajraoui et al., 2017; Wagner et al., 2021)
- ▶ Forward model based on a 1D diffusion equation solved with the finite element method, considering a truncated Karhunen-Loève expansion with  $d = 62$  modes, with coefficients  $X_1, \dots, X_d$  endowed with a standard normal prior  $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$



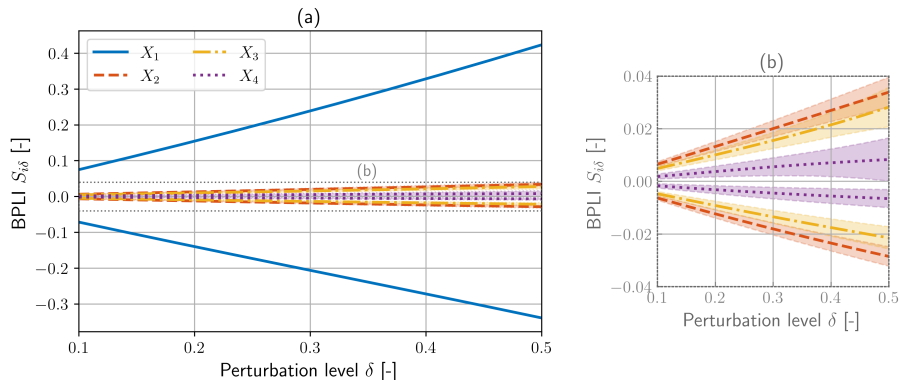
*Few realizations of diffusion coefficient*



*Output concentration associated to diffusion coefficient realizations*

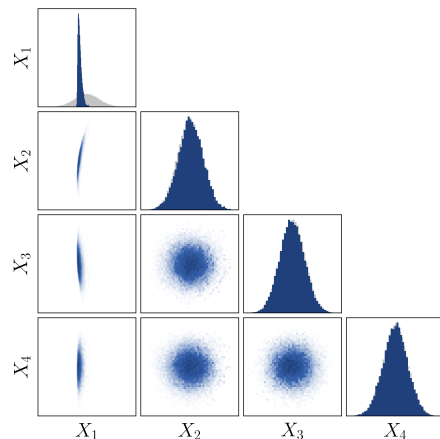
# Case of a high-dimensional diffusion inverse problem

- Computation of minimum/maximum BPLI over the Fisher sphere, for few Karhunen-Loève modes:



(a) full view; (b) zoomed view (detail of  $S_{2\delta}, \dots, S_{4\delta}$ )

# Case of a high-dimensional diffusion inverse problem



*Comparison of reference prior and posterior 1D and 2D marginals of field coefficients  $(X_1, \dots, X_4)$*

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